

Action Sequence Analysis Using Temporal Commonsense Knowledge

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Commonsense temporal action knowledge

- Temporal commonsense reasoning:
 - Understanding the temporal context of phrases, actions, and events to solve problems.
- Innate ability possessed by humans, e.g:
 - "drink a cup of coffee" usually performed over a longer duration than "drink a glass of water"
- Essential for temporal natural language processing tasks.
- Applications:
 - Timeline summarization.
 - Temporal question answering.
 - Temporal natural language inference.
 - Automated planning systems, e.g robotics.

Motivation

- Although maybe LLMs = near perfect at this, applications may require deployment on edge devices with limited computational resources.
- Compact models (e.g., TinyBERT) offer a promising trade-off between performance and energy use.
- Evaluate models for predicting temporal action properties and identifying anomalies in sequences.
- Therefore we experiment on fine-tuning smaller models using a dataset encapsulating commonsense temporal information.

Dataset Origin

wikiHow

- Extract: title, headlines, text.
- Headlines associated with each step generally describe actions in the form of short, concise sentences.

How to Tie a Ribbon Around a Box

Co-authored by [Amy Guerrero](#)

Last Updated: October 23, 2022 [References](#)

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METHODS

- 1 Tying a Basic Bow
- 2 Tying Ribbon Diagonally
- 3 Creating a Woven Look

OTHER SECTIONS

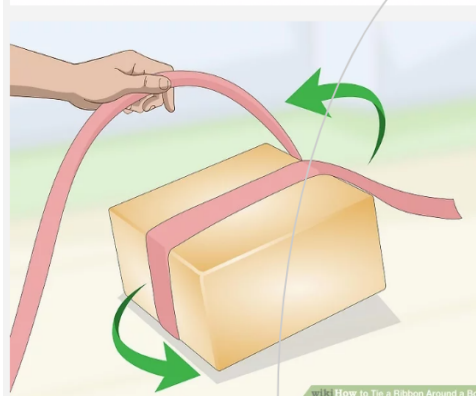
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You've got your gift all wrapped up and ready to go. All it needs is a pretty bow—and tying a ribbon bow around the box will give you a nicer, fancier feel than using a stick-on bow. We'll show you how to tie a basic ribbon bow as well as how to create a diagonal bow and a fancy woven look around your gift box!

1 Drape the ribbon horizontally across the top of the box. Leave about 4 to 8 inches (10.16 to 20.32 centimeters) hanging off the side for the tail of the bow. Do not cut the ribbon yet.^[1]

- It's better to leave more ribbon hanging off the side than too little. You can always cut it shorter later.

ADVERTISEMENT



2 Pull the rest of the ribbon under the box and back to the front. Don't flip the box over, or you may lose your place. Instead, lift the box up and bring the rest of the ribbon behind it. Set the box back down once the ribbon comes out the other side.^[2]

Action sequences

Example:

Title: **How to Tie a Ribbon Around a Box**

Action 1: Drape the ribbon horizontally across the top of the box.

Action 2: Pull the rest of the ribbon under the box and back to the front.

Action 3: Cross the ribbons across the front of the box.

Action 4: Wrap the ribbon around the back of the box and back to the front.

Action 5: Measure the ribbon against the first piece and cut it.

Action 6: Adjust the bow, then trim off the excess ribbon.

(in this example, *length* = 6)

Crowdsourced CoTAK dataset

- Created in our previous work.
Extended for this study.
- Used Amazon Mechanical Turk
 - The annotation process involved presenting every action sequence extracted from WikiHow.
- We asked the workers to think about each action and the dataset constructed using majority voting.

Temporal annotation

amazon

mechanical turk

Each of these sentences relate to **How to Take Care of Outdoor Exotic Animals**. C

1. Make a living space for them.

How long does the action take to perform?

• ☐ Seconds ☐ Minutes ☐ Hours ☐ Days ☐ Months or longer

How long do the effects of the action last?

• ☐ Seconds ☐ Minutes ☐ Hours ☐ Days ☐ Months or longer

2. Get feeding supplies.

How long does the action take to perform?

• ☐ Seconds ☐ Minutes ☐ Hours ☐ Days ☐ Months or longer

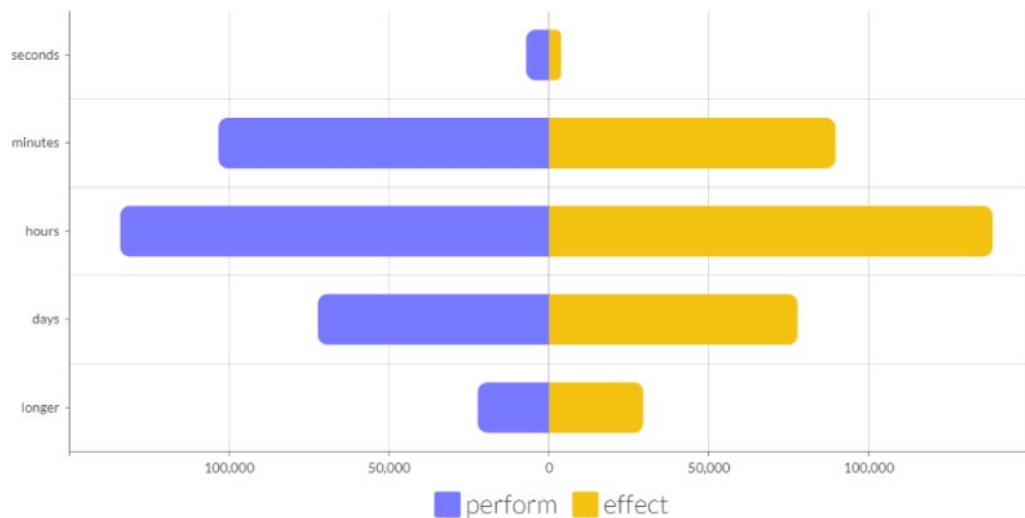


Each sequence sent to 3 workers

```
{ "title": "How to Tie a Ribbon Around a Box",
  "len": 6,
  "actions": [
    {
      "description": "Drape the ribbon horizontally across the top of the box.",
      "duration": {
        "period": "seconds",
        "confidence": 1
      },
      "effect": {
        "period": "minutes",
        "confidence": 3
      }
    },
    {
      "description": "Pull the rest of the ribbon under the box and back to the front.",
      "duration": {
        "period": "seconds",
        "confidence": 3
      },
      "effect": {
        "period": "seconds",
        "confidence": 2
      }
    }
  ]
}
```

Dataset characteristics

- 52,136 action sequences and in total 339,184 actions annotated with both perform and effect duration temporal information.



- Temporal categories:
 - *seconds, minutes, hours, days, longer*
- Perform duration
- Effect duration
- For this work we also added “occurrence type” which can be “singular” or periodic.

Experiments

With the aim of comparing a range of models, we experimented with fine-tuning of:

Model	Params (M)	Type	Suitability
SmolLM2	135	Lightweight	Edge/mobile
RoBERTa-Base	125	Full Model	Server/cloud
TinyBERT	14	Distilled	Edge/mobile/embedded
DistilBERT	66	Distilled	Edge/server

Prediction of temporal characteristics

- Perform
 - How long the action takes to perform.
- Effect
 - How long the effects of the action last for.
- Occurrence Type
 - Singular or Periodic

Results

Perform Period Prediction			
Model	Dataset	Acc (F1)	
		Step only	Goal + Step
SmolLM2	high-confidence	0.64 (0.63)	0.64 (0.63)
	All	0.76 (0.74)	0.72 (0.70)
TinyBERT	high-confidence	0.63 (0.61)	0.7 (0.68)
	All	0.7 (0.69)	0.7 (0.69)
DistilBERT	high-confidence	0.65 (0.64)	0.65 (0.63)
	All	0.71 (0.69)	0.72 (0.71)
RoBERTa-Base	high-confidence	0.66 (0.64)	0.74 (0.73)
	All	0.74 (0.72)	0.74 (0.71)

Results

Effect Period Prediction			
Model	Dataset	Acc (F1)	
		Step only	Goal + Step
SmolLM2	high-confidence	0.55 (0.53)	0.57 (0.54)
	All	0.64 (0.62)	0.61 (0.6)
TinyBERT	high-confidence	0.53 (0.52)	0.57 (0.56)
	All	0.59 (0.57)	0.58 (0.56)
DistilBERT	high-confidence	0.55 (0.53)	0.56 (0.55)
	All	0.59 (0.58)	0.61 (0.6)
RoBERTa-Base	high-confidence	0.57 (0.55)	0.56 (0.55)
	All	0.61 (0.6)	0.61 (0.6)

Results

Occurrence Type Prediction		
Model	Dataset	Acc (F1) Goal + Step
SmolLM2	high-confidence	0.78 (0.8)
	All	0.79 (0.81)
TinyBERT	high-confidence	0.72 (0.75)
	All	0.75 (0.78)
DistilBERT	high-confidence	0.8 (0.78)
	All	0.82 (0.81)
RoBERTa-Base	high-confidence	0.8 (0.82)
	All	0.82 (0.84)

Action sequence prediction

- Goal/Step Alignment:
 - Binary classification task to determine if a given action (step) matches the intended goal.
- Step Position Prediction:
 - Multiclass classification (first, middle, last) to predict where an action occurs in its sequence.

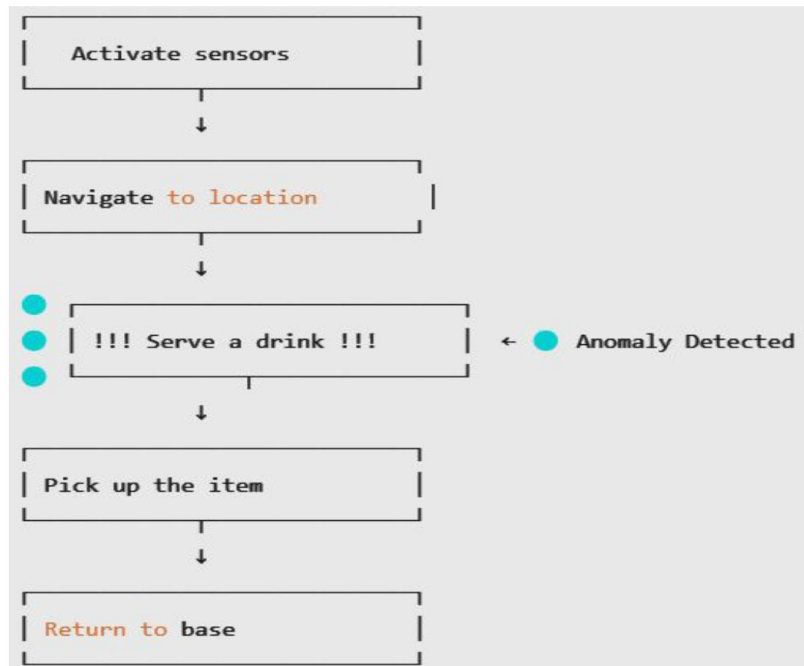
Results

Goal/Step Alignment		
Metric	Acc (F1)	
	Basic	Enhanced
SmolLM2	0.89 (0.89)	0.9 (0.9)
TinyBERT	0.82 (0.81)	0.82 (0.81)
DistilBERT	0.89 (0.87)	0.88 (0.85)
RoBERTa	0.9 (0.9)	0.9 (0.9)

Step Position Prediction		
Metric	Acc (F1)	
	Basic	Enhanced
SmolLM2	0.61 (0.58)	0.62 (0.62)
TinyBERT	0.69 (0.61)	0.69 (0.61)
DistilBERT	0.7 (0.62)	0.69 (0.63)
RoBERTa	0.7 (0.62)	0.7 (0.65)

Anomaly detection

- Used TinyBERT, optimized for low-resource settings.
- Defined four anomaly types:
 - **Goalstep Anomaly**: Wrong step but correct duration
 - **Temporal Anomaly**: Correct step but wrong duration
- Anomalies injected into sequences to simulate real-world deviations.



Results

- Goalstep anomaly prediction achieved high accuracy: $F1 = 82.2$
- Temporal anomalies harder to detect, not better than random guess.
- However this is for individual actions in a sequence, therefore detecting at least one anomaly (if present) in a sequence remains high.
- Demonstrated feasibility of low-power anomaly detection using compact models.

Conclusions and future work

- Lightweight models like TinyBERT and SmolLM2 can effectively predict temporal commonsense properties (duration, recurrence).
- These models also perform goal-step alignment and sequence anomaly detection with competitive accuracy, despite limited model size.
- Future work will be to evaluate models in real planning environments (e.g., robotics).